

ROHAN SINGH

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EDUCATION

Columbia University - School of Engineering and Applied Science New York, NY
Bachelor of Science in Computer Science, Minor in Religion Expected: 2027

- **Awards:** [Egleston Scholar](#) (Awarded to the top 1% of Columbia Engineering, \$10,000 Stipend), Dean's List
- **Coursework:** Data Structures, Intro to Databases, Linear Algebra & Probability, Multivariable Calculus (Calc 3 & 4)
- **Clubs:** Columbia Venture Partners, Columbia Raas dance (Production Chair)

EXPERIENCE

Acadia Pharmaceuticals Inc. June 2025 – August 2025
Software & AI Intern - Talent Acquisition Operations San Diego, CA

- Deployed a Hiring Hub with JavaScript features that improved resource lookup time for **30+** hiring managers.
- Constructed a RAG chatbot to answer hiring questions using a Pinecone vector database for efficient search.
- Reached **87%** chatbot response accuracy by collaborating with DEIB and HR teams to collect feedback.

Columbia Engineering Systems Software Lab - ([Github](#)) Sept. 2024 – April 2025
Machine Learning and Software Research Assistant New York, NY

- Developed ACAI, an automated testing platform using Python to detect LLM chatbot hallucinations.
- Optimized document topic extraction accuracy by **30%** using asynchronous scripts for chatbot content evaluation.
- Publishing paper as a 2nd author and integrated REST APIs to benchmark PDF extraction methods.

University of Arkansas SEEDS Cybersecurity Center - ([Github](#)) May 2023 – May 2024
Cybersecurity and Artificial Intelligence Research Assistant Fayetteville, AR

- Identified data trackers by **91%** with an OpenAI API-based Chrome extension using HTML, CSS, and JavaScript.
- Placed **2nd** at the 2024 Congressional App Challenge and received recognition from state senator.
- Presented at the IEEE GreenTech 2024 Conference and submitted to be published in Frontiers of Big Data.

Walmart Global Tech Dec. 2022 – May 2023
Software Engineering and Computer Vision Intern Bentonville, AR

- Created a mobile QR-code scanner app using Java/API and React Native, accelerating workflow in **5** pilot sites.
- Implemented CI/CD pipelines for automated testing, increasing deployment speed by **25%**.
- Collaborated with HR and sites for feedback to align app features with operational needs and enhance UI/UX.

PROJECTS & PROGRAMS

Jane Street First Year Trading & Technology Program March 2025

- Completed hands-on probability and systems engineering challenges with Jane Street engineers.

HackPrinceton Hackathon - AI Agentic News Feed | *Node.js, LangChain, PostgreSQL* November 2025 – Present

- Scraped local government documents with LangChain AI agents to create a curated news feed. ([View Demo](#))

Novel Artificial Pancreas - ([Project](#)) | *Python, MySQL, Swift, Amazon Web Services (AWS) IoT* Jan. 2023 – Aug. 2023

- Invented a novel artificial pancreas with an AI time-series model, an AWS IoT server, and a MySQL database.
- Patent-pending, published paper, and recognized by the United States Air Force and Regeneron Pharmaceuticals.

Tuberculosis Detection using AI - ([Project Write-Up](#)) | *Python, Docker, Git* Dec. 2022 – July 2023

- Built deep learning model using TensorFlow, Keras, and scikit-learn to predict tuberculosis using cough audio.
- Attained model accuracy of **82%**, ranked **2nd** in the challenge and received authorship in the final manuscript.

PATENTS & PUBLICATIONS

U.S. Patent Application No. 63/602,977 - Sole Inventor for a novel artificial pancreas system

IEEE GreenTech 2024 - First Author - [TraceMonitor: A Novel AI-Based Chrome Extension to Enhance User Awareness...](#)

Oxford Academic - Consortium Author - [Accelerating cough-based algorithms for pulmonary tuberculosis screening...](#)

bioRxiv - First Author - [Optimizing Glycemic Control in Type 1 Diabetic Patients using an AI-Based Artificial Pancreas...](#)

TECHNICAL SKILLS & INTERESTS

Languages: Java, Python, JavaScript, SQL, Swift, HTML, CSS

Tools: Amazon Web Services (AWS) IoT, Node.js, MySQL, CI/CD, Firebase, Git, Docker, Jira

Libraries & Frameworks: React Native, TensorFlow, REST APIs, pandas, Keras, scikit-learn, NumPy, Axios, PyTorch

Awards: 4th Place Grand Award at the International Science & Engineering Fair, National Merit Finalist